# Mean, Median, and Baselines

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

## Teach a baseline result with both datasets visible.

Supply every tournament value,explain the even-count median and separate sensitivity from source truth or general performance.

**Keep one habit:** Require training-only fitting,per-case errors,units and bounded comparison claims.

MAIN LESSON · SECTION 1

## Prepare the connected experiment

> Keep prior exposure, shared practice and independent evidence distinct.

### Goal and main teaching plan

Primary target: Fit **mean** and **median constant baseline**s on **training** targets

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

## Materials and prerequisite

Teach two 45-minute sessions. This week focuses on constant **prediction** baselines. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Opposite signs can cancel; a magnitude-based loss avoids that cancellation.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Optional Python 3 editor; no external packages or AI accounts required

## Opening retrieval

Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish numeric **prediction** from classification

Reference example, not the only acceptable wording: Teacher explanation: Residuals -1, 1, 0 cm; the first record is overpredicted.

Without opening the old lesson, explain one key idea from Precision and Recall. Give an example and a mistake that the idea helps you avoid.

Look for: Choose the denominator that answers the question

Reference example, not the only acceptable wording: Teacher explanation: Predicted positives TP+FP=7; reference positives TP+FN=4.

Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.

Look for: Interpret calibration using groups of forecasts

Reference example, not the only acceptable wording: Teacher explanation: 2/10=.2 and 6/10=.6.

## End goal

By the end, I can explain **mean**, **median**, and baselines using a traced example and a stated limitation.

## ▼ Earlier wording, alternatives and picture transcripts

### Teach the learning cycle

### Retrieval answer guidance

### Regression and Residuals

### Precision and Recall

### Scores and Calibration Intuition

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

### What the learner must demonstrate

For **training** targets [1,3,11], compute **mean** and **median constant baseline**s. Compare total absolute **training** error for each. Where must final performance be measured?

### Concept to explain

**Mean** and **median** are transparent regression baselines; **outlier**s affect them differently.

| Term | Meaning |
| --- | --- |
| mean | sum divided by count |
| median | middle ordered value |
| outlier | value far from most observations |
| location baseline | single typical value predicted for all cases |

### Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

**Preparation for the connected explanation**

**Readiness answer**

**End goal for the pictorial case**

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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MAIN LESSON · SECTION 2

# Teach fit and evaluate as separate target sets

> Include the sorted even-count median and the actual per-case errors.

## Constant-fitting contract and source

Before a complex regression model, fit a simple constant predictor. A mean baseline uses the training average. Sort the training targets for a median: use the middle value for an odd count, or the average of the two middle values for an even count. The methods ignore query features but provide a reproducible reference. Evaluation targets must not determine the constants.

Before a complex regression model, fit a simple constant predictor. A **mean** baseline uses the **training** average. Sort the **training** targets for a **median**: use the middle value for an odd count, or the average of the two middle values for an even count. The methods ignore query features but provide a reproducible reference. Evaluation targets must not determine the constants.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines constant **prediction** baselines.

## Trace main fitted values and equal MAEs

**Training** targets [2, 4, 6, 12] have **mean** 24/4=6 and **median** (4+6)/2=5. On separate targets [4, 8], the **mean** baseline predicts [6, 6] with **absolute error**s [2, 2], averaging 2.

The **median** baseline predicts [5, 5] with errors [1, 3], also averaging 2.

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

| Term | Meaning |
| --- | --- |
| Constant baseline | One fixed prediction for every query |
| Training mean | Sum of training targets divided by their count |
| Training median | Middle value of sorted training targets |
| Absolute error | Magnitude of observed minus predicted |

### Constant prediction baselines at a glance

| PART | EVIDENCE OR ACTION | INTERPRETATION |
| --- | --- | --- |
| Training targets | Fit mean 6 | Freeze constant |
| Training targets | Fit median 5 | Freeze constant |
| Evaluation targets | Compare fixed predictions | Both MAE 2 here |

Mean and median answer different training objectives. A mean minimizes total squared error among constants; a median minimizes total absolute error. This does not guarantee either wins on a new evaluation set. With extreme values, the mean can move much more than the median.

**Calculate MAE for both baselines on [4, 8]. Explain the steps or evidence behind your response.**

## Read separate pictured training/evaluation values

Fit two constant height baselines and compare them on held-out examples.

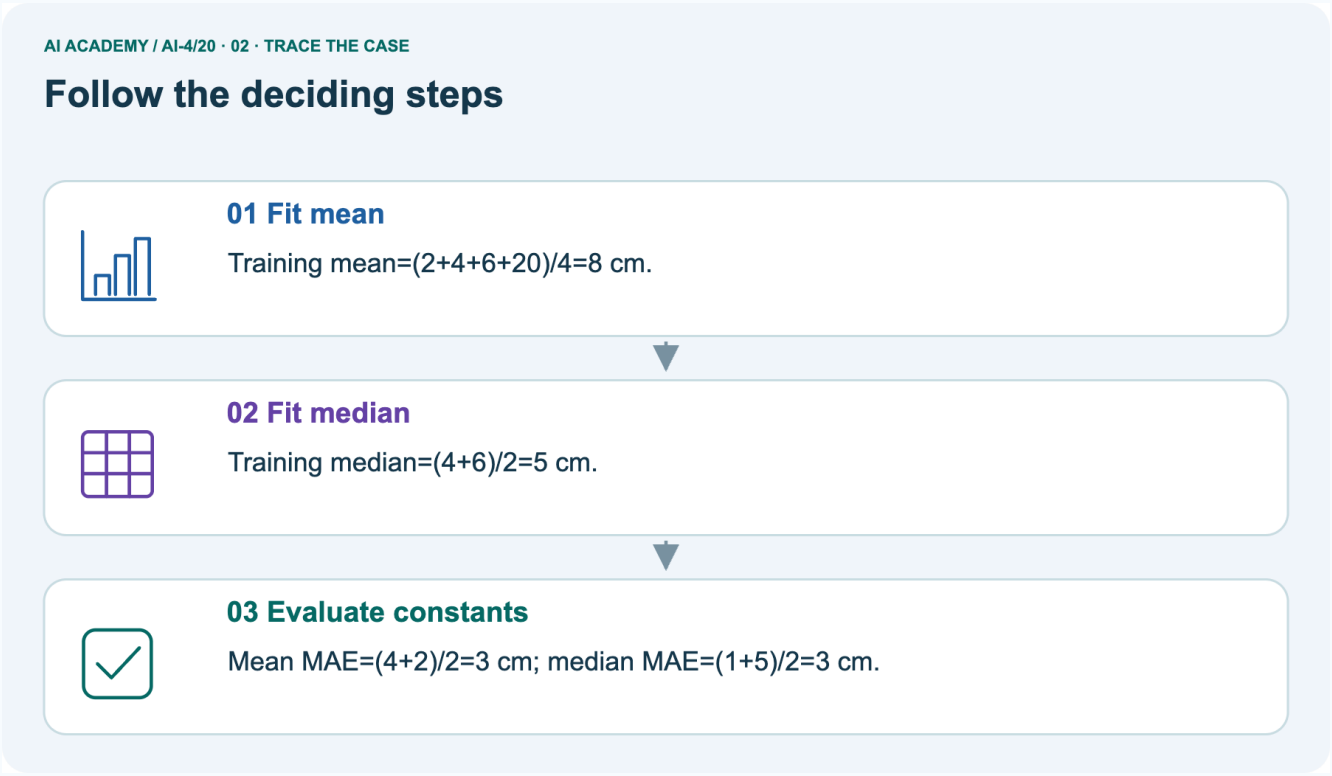
**Training** heights [2, 4, 6, 20] cm. Evaluation heights [4, 10] cm. Fit constants only from **training**.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Main training2,4,6,12gives mean6and median5;validation4,8gives errors[2,2]and[1,3],both MAE2cm. Picture training2,4,6,20gives mean8and median5;evaluation4,10gives errors[4,2]and[1,5],both MAE3cm. Replacing only pictured training20with8gives mean5and median5,both still evaluation MAE3cm. Keep the evaluation targets fixed.

This supplied change demonstrates sensitivity,not evidence that the extreme value was incorrect or permission to choose training edits from evaluation results.

## Trace constants,held-out tie and training-value change




Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Fit mean | Training mean=(2+4+6+20)/4=8 cm. |
| 2. Fit median | Training median=(4+6)/2=5 cm. |
| 3. Evaluate constants | Mean MAE=(4+2)/2=3 cm; median MAE=(1+5)/2=3 cm. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

**Mean** and **median** have different loss properties, but their held-out performance depends on the particular evaluation cases.

Change: Replace **training** 20 with 8 while keeping evaluation unchanged.

**Prediction**: Both fitted constants become5 cm and both evaluation MAEs remain3 cm.

Reason: **Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

### ▼ Additional2,4,12training example

**Training** heights are 2,4,12 cm. Compare constant **mean** and **median** baselines.

Compute both and **training absolute error**s.

**Mean**=6, **median**=4. **Mean** errors 4,2,6 total12; **median** errors 2,0,8 total10. **Median** minimises **absolute error** here.

### ▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Choose and justify a baseline under an extreme value.

MODEL IDEA **Mean** and **median** are transparent regression baselines; **outlier**s affect them differently.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Heights are 10, 11, 12, 13, and 54 cm. **Mean** = 100 ÷ 5 = 20 cm.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

STEP 1 Heights are 10, 11, 12, 13, and 54 cm.

STEP 2 **Mean** = 100 ÷ 5 = 20 cm.

STEP 3 **Median** is 12 cm, the middle ordered value.

STEP 4 For typical-height **absolute error**, **median** is more robust; still report why 54 is present.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

### Concrete teaching case

### Exact explanation to model

### Explain the added worked example

### Key terms

### Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

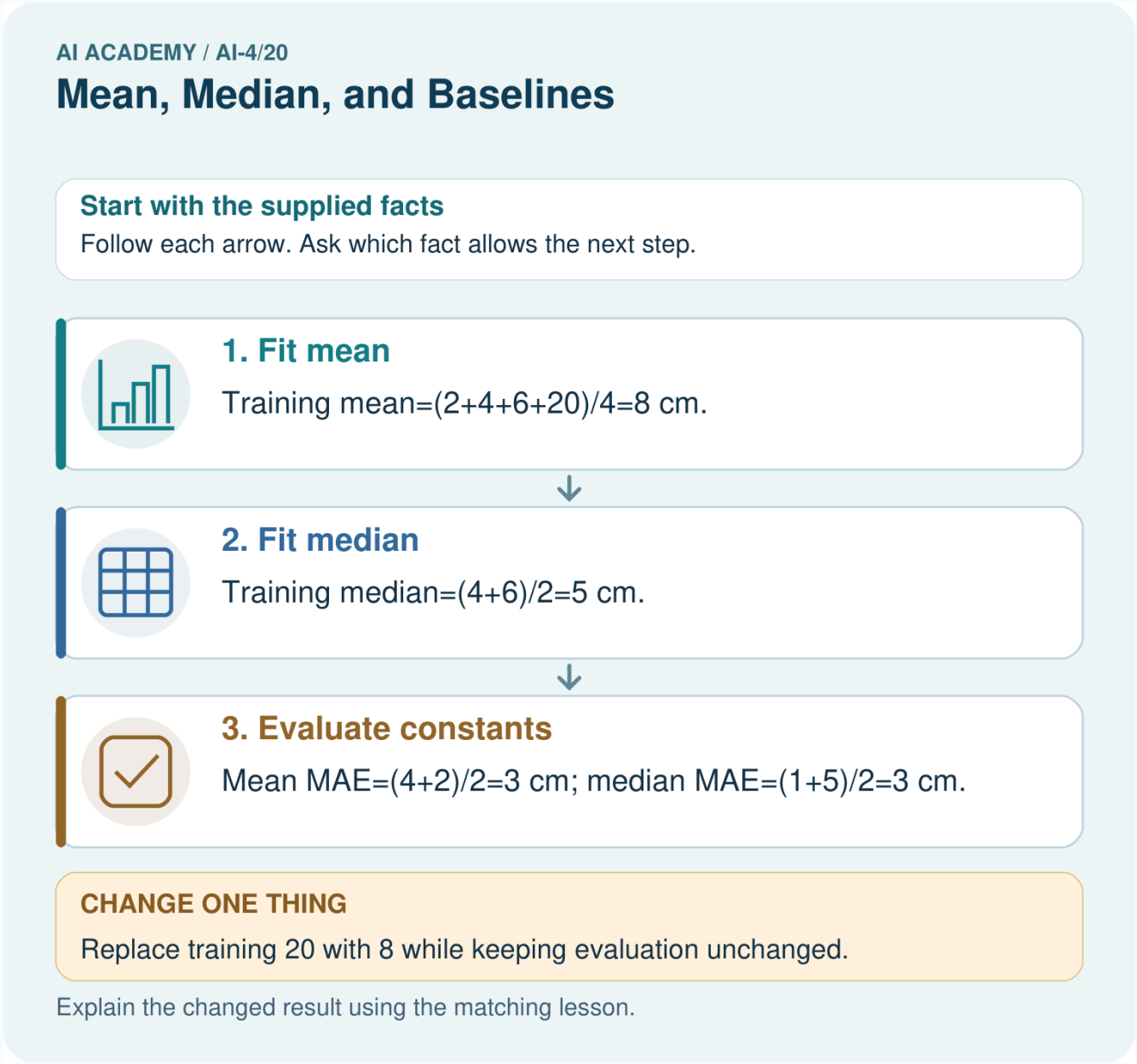
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

| Thinking move | Teacher prompt |
| --- | --- |
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

## A pictorial worked case: Mean, Median, and Baselines

### The facts we will use




Follow the labelled picture with the matching explanation. Open the vector diagram

**Why this result follows**

**What this example does not establish**

**Change one thing in the pictured case**

**Ask while showing the picture**

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

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MAIN LESSON · SECTION 3

# Guide the calculation, then remove help

> Collect each learner's prediction before showing its key.

## Current completion task and checked key

Completion task: Calculate MAE for both baselines on [4, 8].

Teacher check: Teacher explanation: Both score 2; their per-record **absolute error**s differ.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

## Individual misconception check

Use the worked case for Constant **prediction** baselines. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: **Mean** 6; **median** 5.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

## Pictorial question and changed-case reasoning

Does either original baseline win under evaluation MAE? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

No, both have3 cm MAE on the supplied evaluation set; do not claim a strict winner from the **training** statistic alone. Changed-case check: Both fitted constants become5 cm and both evaluation MAEs remain3 cm.

**Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

Changed-case reasoning: Both fitted constants become5 cm and both evaluation MAEs remain3 cm. **Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

## Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

## ▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

### Board sequence

Write the given facts: **Training** heights are 2,4,12 cm. Compare constant **mean** and **median** baselines.

Write the question to explain: Compute both and **training absolute error**s.

Reveal after **prediction**s: **Mean**=6, **median**=4. **Mean** errors 4,2,6 total12; **median** errors 2,0,8 total10. **Median** minimises **absolute error** here.

Use the earlier transfer question as additional practice. Reserve the selected fresh **training**1,3,5,15and evaluation3,9task for the first individual attempt and record support used.

### Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

### Model these guided questions

Guided response: Both score 2; their per-record **absolute error**s differ. Exit response: It fits using evaluation answers and makes the baseline unavailable for genuinely unseen cases.

### Case-specific teaching sequence

Stage · Teacher action

> **Stage:** Session 1 start to 7 minutes
>
> **Teacher action:** Use the prerequisite question: Why can signed residuals hide error size? Check the explanation before proceeding.

**Stage:** 7 to 18 minutes

**Teacher action:** Read the opening and model the worked evidence. Ask students to identify constant baseline in context.

**Stage:** 18 to 32 minutes

**Teacher action:** Complete W 1 and W 2 with support: fit two constants; score separately. Require the evidence, not only the final response.

**Stage:** 32 to 45 minutes

**Teacher action:** Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Trace the Python check after the paper calculation.

**Stage:** Session 2 start to 8 minutes

**Teacher action:** Retrieve the distinction between constant baseline and training mean. Invite a counterexample to an overconfident claim.

**Stage:** 8 to 20 minutes

**Teacher action:** Discuss the application and complete W 3: Replace 12 by 40. What changes?

**Stage:** 20 to 35 minutes

**Teacher action:** Collect W4 independently before giving hints. Assess these outcomes: Fit mean and median constant baselines on training targets; Compare constant predictions on separate records. Record any support used.

**Stage:** 35 to 45 minutes

**Teacher action:** Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

**Reduce help deliberately**

**Individual reasoning check**

**Pictorial practice question**

**Reasoned teacher answer**

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MAIN LESSON · SECTION 4

# Run the complete supplied investigation

> Keep predictions, observed or supplied outcomes and unmeasured cases separate.

## Supplied inputs and checked guidance

Tournament:original constants6/5,validationMAE2/2;changed training40constants13/5,MAE7/2. A single training5makes both5;empty targets unavailable;extreme value needs provenance checking. Independent training1,3,5,15mean6,median4;eval3,9errors[3,3]and[1,5],bothMAE3. Matching held-out scores do not imply equal predictions or general superiority. Preserve source/version and fitting/evaluation roles.

## Predict, investigate and record

1. 1. Write the hypothesis and expected **evidence** before running.
2. 2. Baseline Tournament: compute **mean** and **median**, predict every case, compare MAE, and explain the **outlier**'s effect.
3. 3. Run one normal, one boundary, one missing, and one shifted case.
4. 4. Calculate the metric from raw counts or row-level errors.
5. 5. Interpret what changed, what did not, and what remains unsafe or unknown.

## Keep the experiment ledger

| Record | What to capture |
| --- | --- |
| Input or evidence | Exact data, instruction or claim |
| Expected result | Answer or safe fallback declared before the run |
| Observed result | Actual result, including errors |
| Explanation | Rule, calculation or evidence explaining the outcome |
| Next test | One justified change and a new prediction |

## Diagnose the current misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: **Mean** 6; **median** 5.

A stronger claim makes a stronger project: It fits using evaluation answers and makes the baseline unavailable for genuinely unseen cases.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: **Mean** 6 and **median** 4. **Mean** errors [3, 3] MAE 3; **median** errors [1, 5] MAE 3.

## Choose accessible support

| What the response shows | Next teaching action |
| --- | --- |
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |

| What the response shows | Next teaching action |
| --- | --- |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

### ▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

#### Make the lab runnable

Use the supplied complete investigation for shared practice. Reserve the selected fresh training1,3,5,15and evaluation3,9task for the individual first attempt before hints or feedback; earlier transfer questions remain additional practice.

#### Diagnose and reteach

MISCONCEPTION TO REPAIR The **mean** is always the fairest and most scientific summary.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

**Address these additional misconceptions**

**Match the support to the first error**

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MAIN LESSON · SECTION 5

# Assess the selected experiment transfer

> Keep this key until after a first attempt; record any help and prior exposure.

### Selected independent task and checked key

Selected case: **Training** [1, 3, 5, 15], evaluation [3, 9]. Fit both constants and calculate MAE.

Teacher explanation: **Mean** 6 and **median** 4. **Mean** errors [3, 3] MAE 3; **median** errors [1, 5] MAE 3.

## Judge trace, scope and support

| Dimension | Evidence required |
|---|---|
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

## Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

## W1 Fit two constants

Find the **mean** and **median** of the worked **training** targets.

Teacher explanation: **Mean** 6; **median** 5.

For the **mean** baseline, sum the worked **training** targets and divide by their count to obtain 6. For the **median** baseline, sort the **training** targets and take the middle value, 5.

Freeze these choices before examining evaluation targets; choosing a new constant from evaluation outcomes would contaminate the comparison.

## W2 Score separately

Calculate MAE for both baselines on [4, 8].

Teacher explanation: Both score 2; their per-record **absolute error**s differ.

For constant 6, **absolute error**s on targets 4 and 8 are 2 and 2, so MAE is (2+2)/2 = 2. For constant 5, the errors are 1 and 3, so MAE is (1+3)/2 = 2.

Equal averages can hide different per-record errors; retain the error table as well as the summary.

## W3 Inspect an extreme value

Replace 12 by 40. What changes?

Teacher explanation: **Mean** becomes 13; **median** stays 5. Investigate validity before deciding treatment.

## W5 Prevent leakage

Why not predict the evaluation **mean** as the baseline?

Teacher explanation: It fits using evaluation answers and makes the baseline unavailable for genuinely unseen cases.

### ▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

#### Worked case answer

**Mean**=6, **median**=4. **Mean** errors 4,2,6 total12; **median** errors 2,0,8 total10. **Median** minimises **absolute error** here.

#### Additional transfer practice question

For **training** heights 2, 3 and 10 cm, calculate **mean** and **median**. Which constant is less influenced by the high value?

#### Reasoned practice answer

**Mean** is 15/3 = 5 cm; **median** is 3 cm. The **median** is less affected by the high value. Fit a baseline from **training** data only.

#### What counts as evidence

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Result | Incorrect or absent | Partly correct | Correct with necessary conditions |
| Reasoning | No supporting trace | Some steps | Clear mechanism, calculation or source support |
| Limit | Overclaims certainty | Vague limitation | Relevant uncertainty or missing fact |
| Transfer | Copies without adapting | Adapts with prompts | Solves a changed case independently |

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

### Feedback

Ask the learner to find the **evidence** that challenges this claim: “**Mean** and **median** must always be equal.” Keep the first attempt and an uncoached retry separate.

### Original transfer answer

**Mean**=5, **median**=3. **Mean** absolute-error total=4+2+6=12; **median** total=2+0+8=10. These are **training** results. Fit the constant using **training** targets and measure final performance on untouched test targets.

### Explanations for the additional workbook examples

### W1 Fit two constants

### W2 Score separately

### W3 Inspect an extreme value

### W4 Apply independently

**Training** [1, 3, 5, 15], evaluation [3, 9]. Fit both constants and calculate MAE.

Teacher explanation: **Mean** 6 and **median** 4. **Mean** errors [3, 3] MAE 3; **median** errors [1, 5] MAE 3.

### W5 Prevent leakage

### Assess independent transfer

### Selected case answer

### Feedback that the learner uses

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MAIN LESSON · SECTION 6

## Connect, support and revisit the experiment

> Retain failures and assess a fresh parallel case after help.

## Project evidence

Document any numeric baseline used in Plant Classifier feature checks, including unit, distribution, and **outlier** decision.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

## Support and selected reassessment

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **constant baseline** and **training mean** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: **Mean** 6 and **median** 4. **Mean** errors [3, 3] MAE 3; **median** errors [1, 5] MAE 3. Fresh retry **training** [2, 2, 8], evaluation [1, 7]: **mean** 4 **median** 2; MAE 3 for each. Require **training**-only **fitting**.

Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

## Delayed return

Suggested return points in this level: ai-4/21, ai-4/23, ai-4/26. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
| --- | --- |
| Learner code and date | |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

## Bridge and reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 21: **Absolute Error** and MAE. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week 21: Absolute Error and MAE. Carry forward the training-derived constants, fixed evaluation targets, per-case absolute errors, units and denominator. Squared loss can be considered as a separate extension with its own units.

### ▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Responsible Plant Classifier

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

## Reference · Alternative planning checklist

> Use alongside the main route without repeating the whole cycle.

### Planning and assessment reference

Teaching target: Fit **mean** and **median constant baseline**s on **training** targets

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Calculate MAE for both baselines on [4, 8].

Reasoned practice answer: Teacher explanation: Both score 2; their per-record **absolute error**s differ.

Check every learner before discussion: Use the worked case for Constant **prediction** baselines. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: **Mean** 6; **median** 5.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Replace **training** 20 with 8 while keeping evaluation unchanged. Require a **prediction** and a reason before comparing. Expected result: Both fitted constants become5 cm and both evaluation MAEs remain3 cm.

Reason: **Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

# Reference · Feedback and evidence records

> Keep supported rehearsal separate from independent performance.

## Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Earlier wording, alternatives and picture transcripts**

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

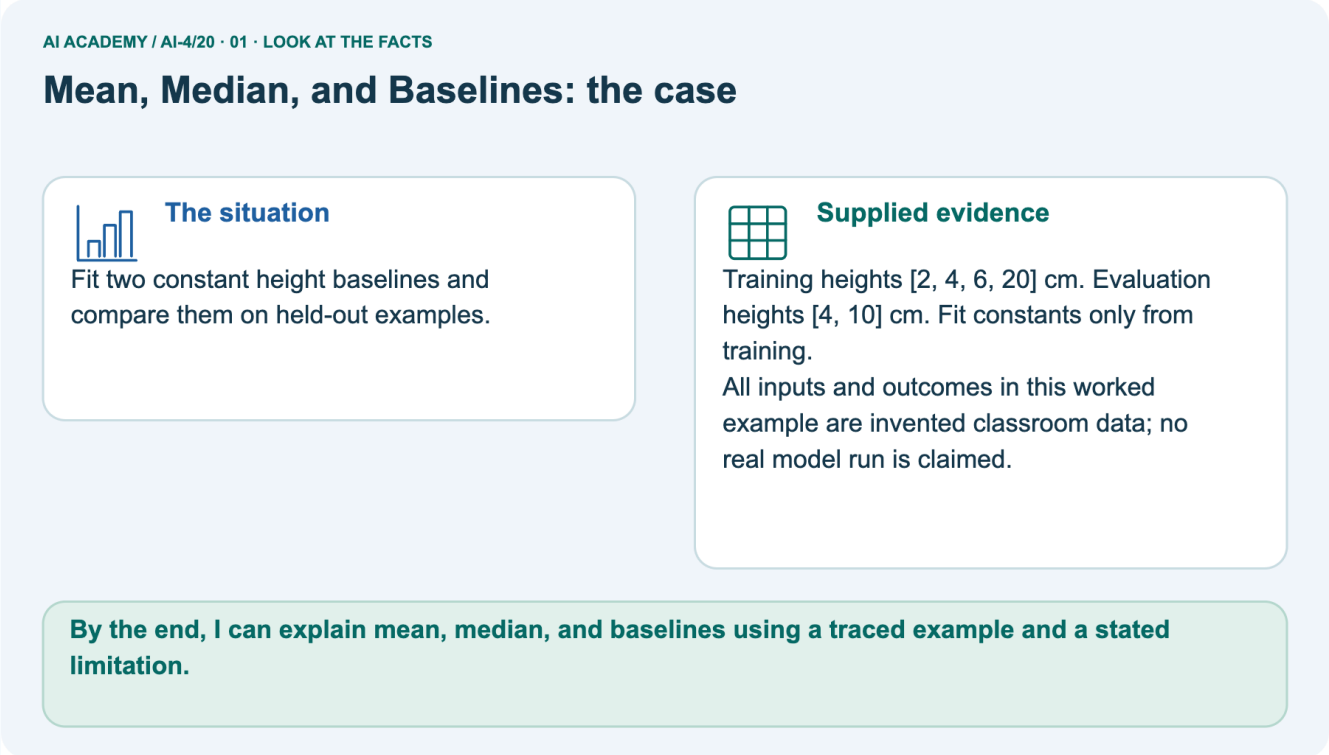
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

## Reference · Teach with the six pictures

> Choose one picture for a specific reasoning move.

## 01. Mean, Median, and Baselines: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

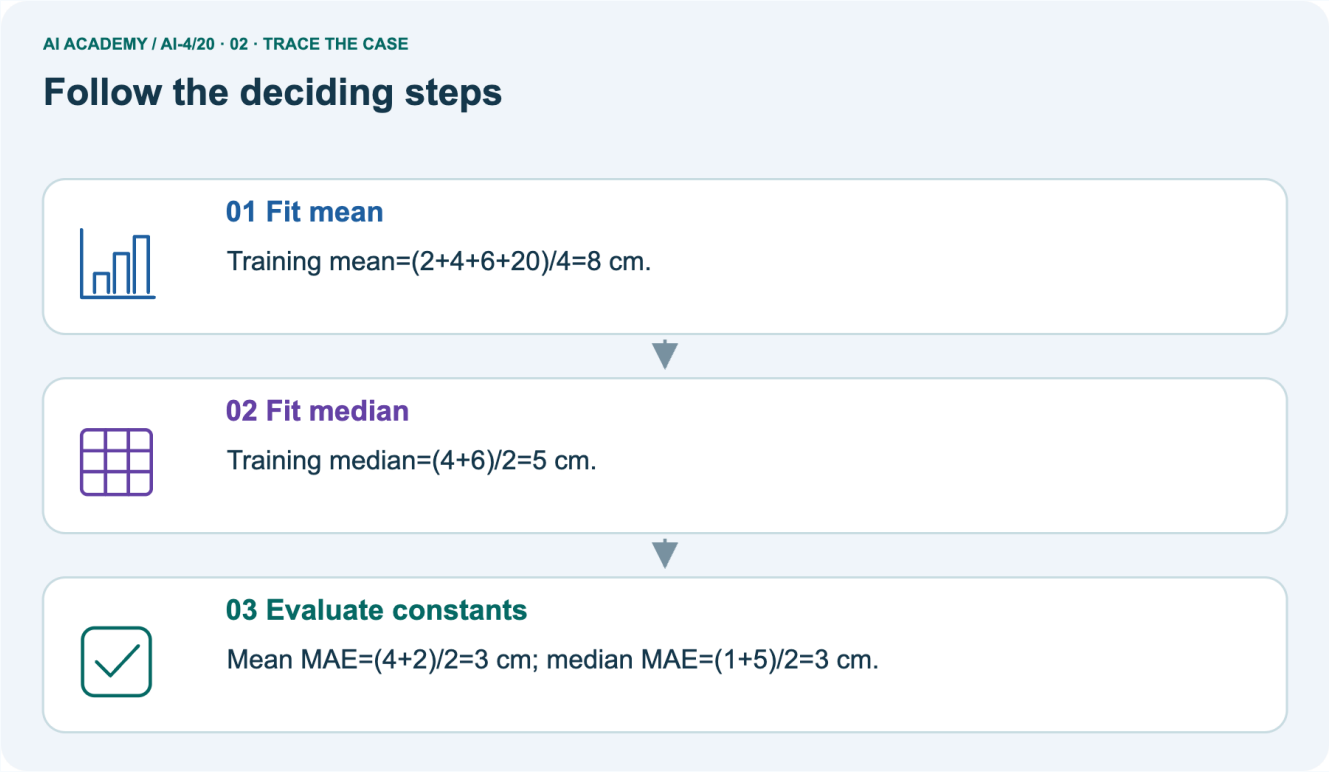
Fit two constant height baselines and compare them on held-out examples. Supplied facts: **Training** heights [2, 4, 6, 20] cm. Evaluation heights [4, 10] cm. Fit constants only from **training**.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 02. Follow the deciding steps



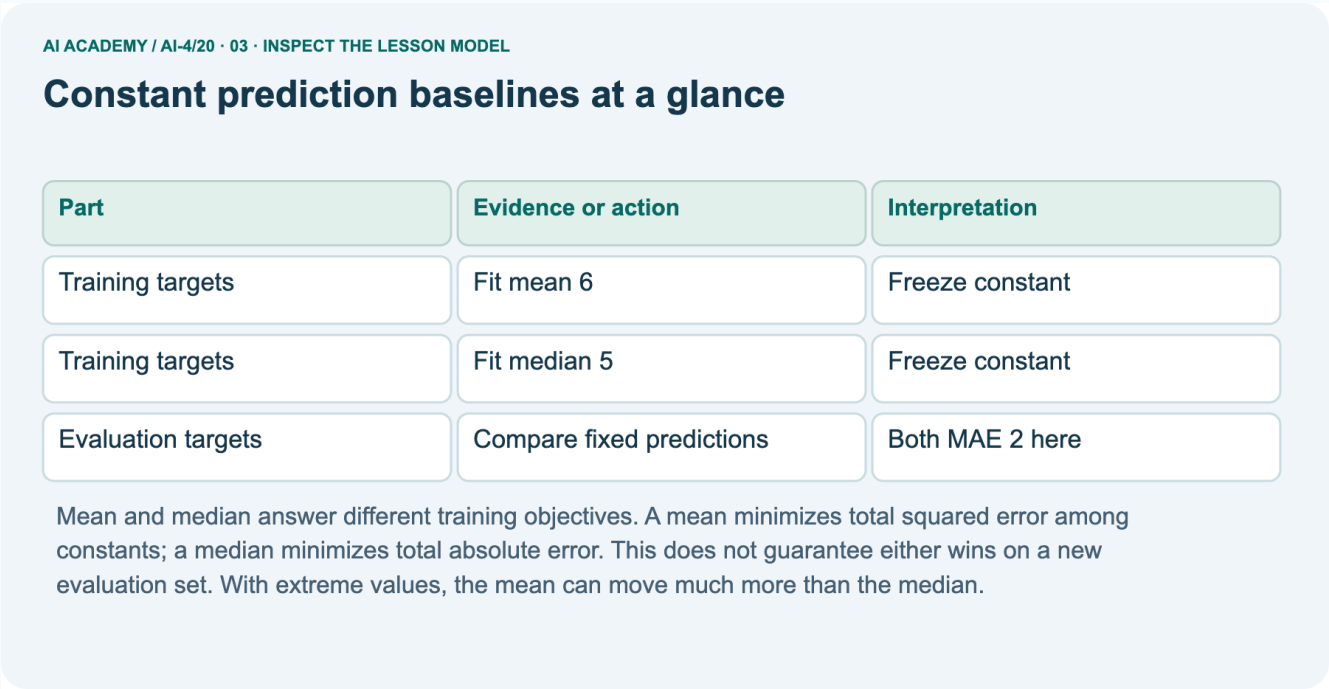

Follow the labelled picture with the matching explanation. Open the vector diagram

Fit **mean**: **Training mean**=(2+4+6+20)/4=8 cm. → Fit **median**: **Training median**=(4+6)/2=5 cm. → Evaluate constants: **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 03. Constant prediction baselines at a glance



AI ACADEMY / AI-4/20 · 03 · INSPECT THE LESSON MODEL

### Constant prediction baselines at a glance

| Part | Evidence or action | Interpretation |
| --- | --- | --- |
| Training targets | Fit mean 6 | Freeze constant |
| Training targets | Fit median 5 | Freeze constant |
| Evaluation targets | Compare fixed predictions | Both MAE 2 here |

Mean and median answer different training objectives. A mean minimizes total squared error among constants; a median minimizes total absolute error. This does not guarantee either wins on a new evaluation set. With extreme values, the mean can move much more than the median.

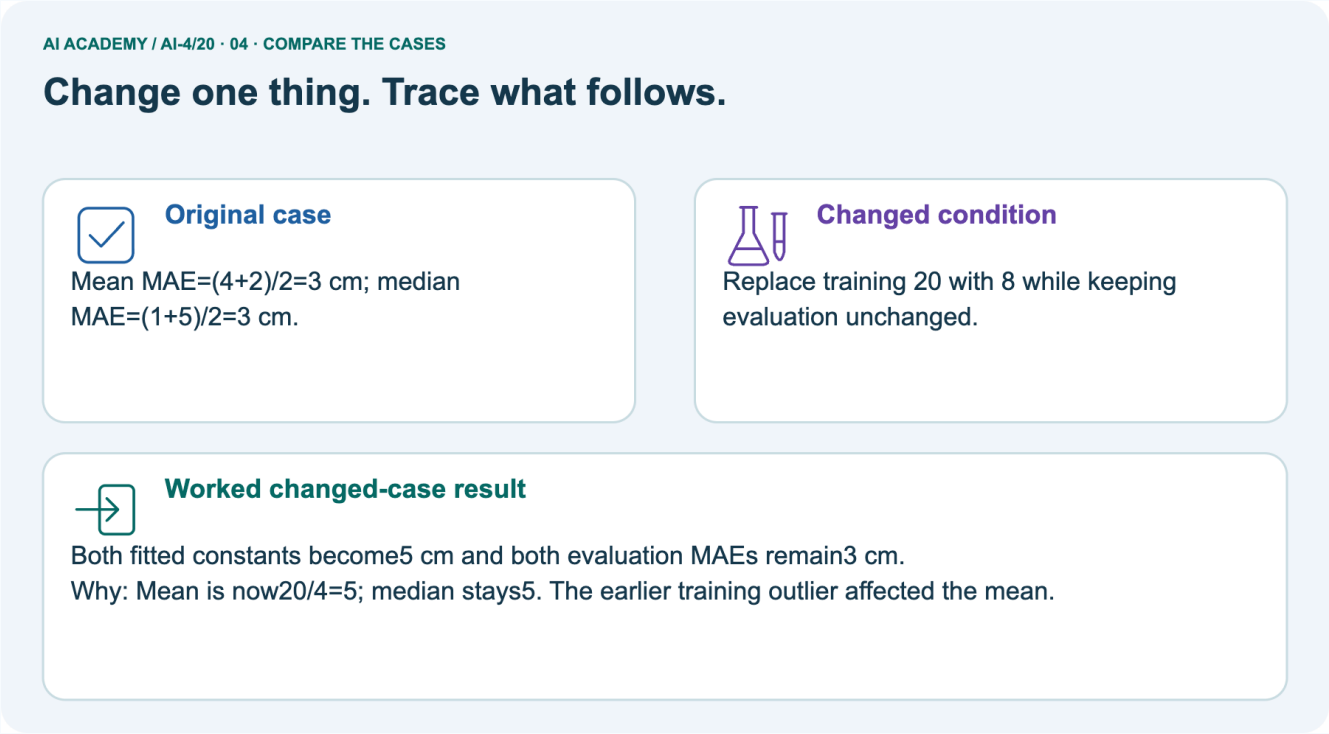
Follow the labelled picture with the matching explanation. Open the vector diagram

Constant **prediction** baselines at a glance. **Training** targets | Fit **mean** 6 | Freeze constant. **Training** targets | Fit **median** 5 | Freeze constant. Evaluation targets | Compare fixed **prediction**s | Both MAE 2 here

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm. Changed condition: Replace **training** 20 with 8 while keeping evaluation unchanged.

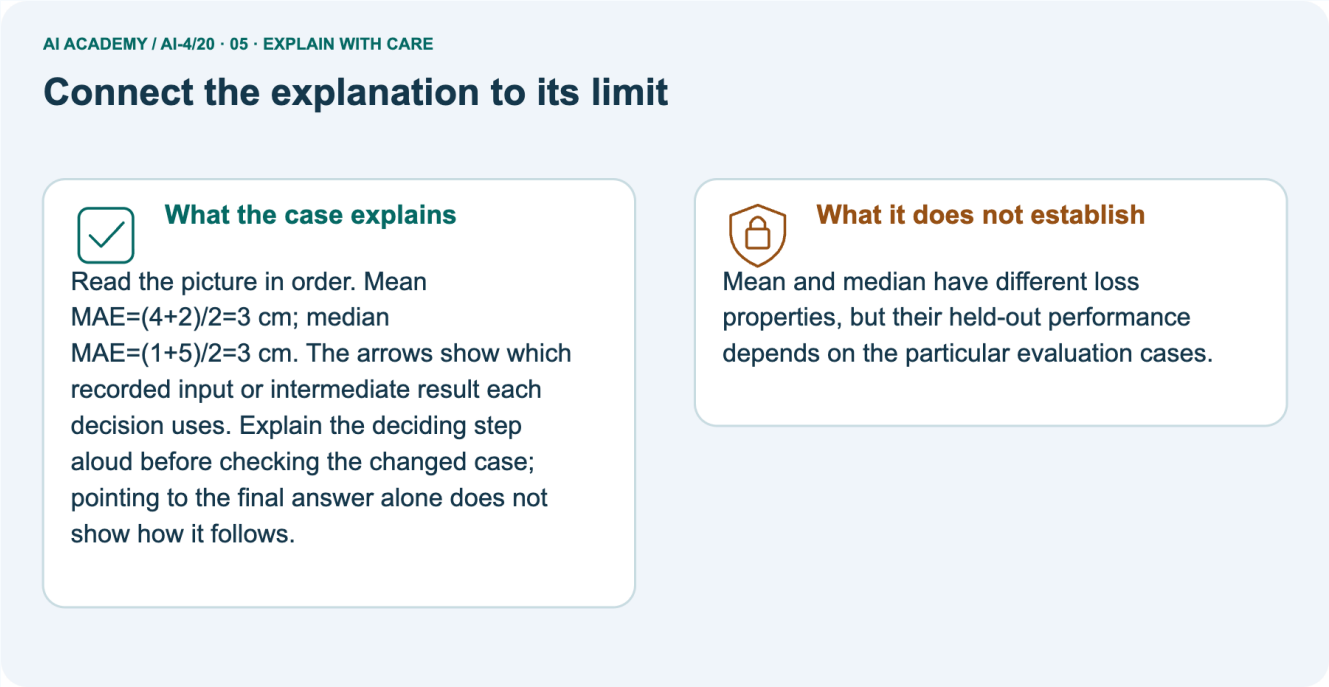
Expected result: Both fitted constants become5 cm and both evaluation MAEs remain3 cm. Reason: **Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: **Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

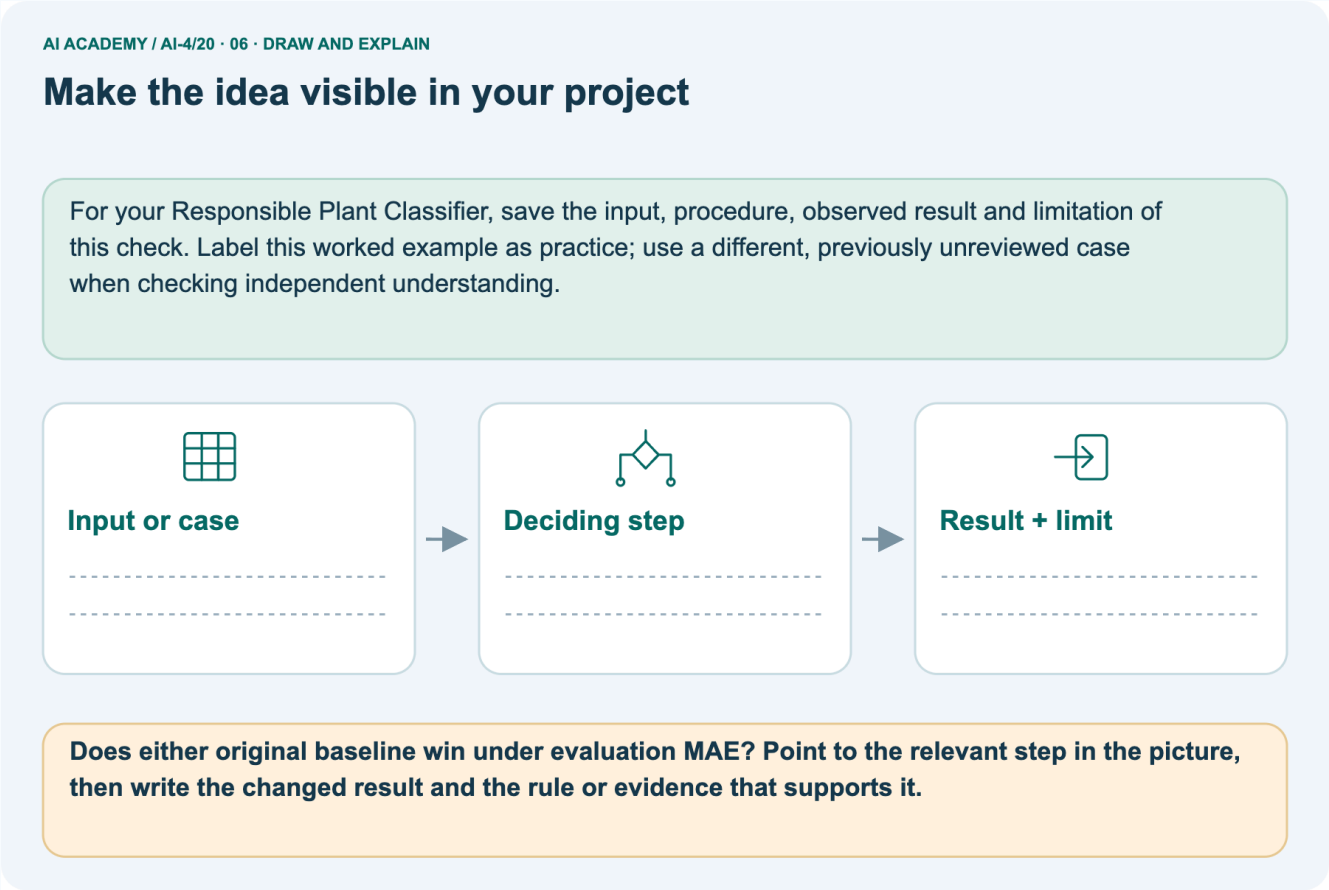
Limit: **Mean** and **median** have different loss properties, but their held-out performance depends on the particular evaluation cases.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: **Mean** and **median** have different loss properties, but their held-out performance depends on the particular evaluation cases.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does either original baseline win under evaluation MAE? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No, both have3 cm MAE on the supplied evaluation set; do not claim a strict winner from the **training** statistic alone.

Changed-case check: Both fitted constants become5 cm and both evaluation MAEs remain3 cm. **Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

## ▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

### 01. Mean, Median, and Baselines: the case

### 02. Follow the deciding steps

### 03. Constant prediction baselines at a glance

### 04. Change one thing. Trace what follows.

### 05. Connect the explanation to its limit

### 06. Make the idea visible in your project

## ▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.